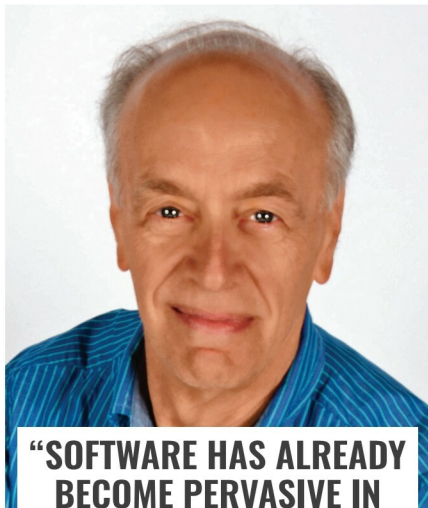




## “SOFTWARE HAS ALREADY BECOME PERVASIVE IN THE DEVELOPED AND THE DEVELOPING WORLDS”

While hardware is acquiring new forms, software is adapting to changes in the landscape. But how do developers benefit from this vicious cycle of hardware and software changes? Likewise, what should decision makers keep in mind while building new technology solutions? American computer scientist **Tony**

**Wasserman** answers these questions in a conversation with **Jagmeet Singh** of *OSFY*. Wasserman also shares his views of the growth prospects of open source, based on his experience as a **board member** of the **Open Source Initiative**, **professor** of the **software management practice** at **Carnegie Mellon, Silicon Valley** and as the **executive director** of the **CMU (Carnegie Mellon University) Center for Open Source Investigation**. Edited excerpts...

Tony Wasserman

### Q How do you see software evolving in the computing world?

Software has already become pervasive in the developed and in the developing worlds. Over the past decade, we have seen important advances in embedded software (now called the Internet of Things) and in data management and analytics, which are being applied to machine learning and artificial intelligence. These advances are leading to increasingly powerful autonomous systems, such as self-driving cars, industrial and personal robots and surveillance systems. Even digital cameras, which are highly computerised, are now able to install additional apps.

While we will still need professional developers to create system level functions and development tools, there will be a growing role for ‘low code’ and ‘no code’ tools. The former will be used by software developers and may involve connecting and/or customising proven software components. The latter will be accessible to people without formal programmer training.

Professional developers will continue to use sophisticated programming languages and development environments, much as they do today, though the preferred languages may evolve over time.

### Q What are the major trends shaping up the IT industry?

The most important trends involve the continuing decrease in the cost of computer hardware, particularly for storage, along with the growth of cloud-based computing services. Consumers have long become accustomed to using hosted services and they make extensive use of mobile devices, Chromebooks and tablets that connect to remote computing services. Enterprises and governments are becoming more aware of the costs of maintaining their own IT systems, and are